Math 208 Midterm 2, Spring 2025 (Sara Billey) — Problem 1

(a) The two columns of A are not multiples of each other (look at row 2 vs row 4), so they are independent: $\boxed{\text{rank}A = 2,\ \text{nullity}A = 2 - 2 = 0}$.

(b) $\text{rank}A^T = \text{rank}A = 2$; A^T has 4 columns, so $\boxed{\text{nullity}A^T = 4 - 2 = 2}$.

(c) $B^2 = \begin{bmatrix} 1 & 16 \\ 0 & 1 \end{bmatrix}$, so $B^2 - B = \begin{bmatrix} 0 & 8 \\ 0 & 0 \end{bmatrix}$ and

$$AB^2 - AB = A(B^2 - B) = \boxed{\begin{bmatrix} 0 & 16 \\ 0 & 0 \\ 0 & 64 \\ 0 & 0 \end{bmatrix}}.$$

(d) It is 4×2, not square, so it is **not invertible**. A nonzero $\mathbf{v}$ with $(AB^2 - AB)\mathbf{v} = \mathbf{0}$: $\boxed{\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}}$.

Math 208 Midterm 2, Spring 2025 (Sara Billey) — Problem 2

Row reduce:

$$C \sim \begin{bmatrix} 1 & -\frac{7}{3} & 0 & 0 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}, \quad \text{pivots in columns } 1, 3, 4.$$

(a) Pivot columns of C: $\left\{ \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} \right\}$ (so $\text{col}C = \mathbb{R}^3$).

(b) The three nonzero rows of the reduced matrix (or the three rows of C, since the rank is 3).

(c) Free variables x_2, x_5, x_6. Clearing fractions:

$$\text{null}(C) = \text{span}\left\{ \begin{bmatrix} 7 \\ 3 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ -1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ -2 \\ -1 \\ 0 \\ 2 \end{bmatrix} \right\} \quad (\text{each checked: } C\mathbf{v} = \mathbf{0}).$$

Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Midterm 2, Spring 2025 (Sara Billey) — Problem 3

(a) Row operations don't change linear relations among columns. The pivot columns of B are 1 and 3, so $\{\mathbf{a}_1, \mathbf{a}_3\}$ is a basis and $\dim \operatorname{col}(A) = 2$. Any set of exactly 2 linearly independent vectors in $\operatorname{col}(A)$ is a basis; sets of a different size, or dependent sets, are not.

(b) From B: $\mathbf{b}_2 = \mathbf{0}$, so $\mathbf{a}_2 = \mathbf{0}$; and $\mathbf{b}_4 = \frac{3}{2}\mathbf{b}_1 - \frac{1}{2}\mathbf{b}_3$, so $\mathbf{a}_4 = \frac{3}{2}\mathbf{a}_1 - \frac{1}{2}\mathbf{a}_3$.

- $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$: **NO**, it contains the zero vector (and has 3 vectors).
- $\{\mathbf{a}_1, \mathbf{a}_3, \mathbf{a}_4\}$: **NO**, three vectors in a 2-dimensional space are dependent.
- $\{\mathbf{a}_1 + 2\mathbf{a}_4,\ \mathbf{a}_3 + \mathbf{a}_4\} = \{4\mathbf{a}_1 - \mathbf{a}_3,\ \frac{3}{2}\mathbf{a}_1 + \frac{1}{2}\mathbf{a}_3\}$: **YES**, two vectors whose coordinates $(4, -1)$, $(\frac{3}{2}, \frac{1}{2})$ have determinant $\frac{7}{2} \neq 0$.
- $\{\mathbf{a}_1,\ \mathbf{a}_3 + 2\mathbf{a}_4\} = \{\mathbf{a}_1,\ 3\mathbf{a}_1\}$: **NO**, they are parallel.

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Math 208 Midterm 2, Spring 2025 (Sara Billey) — Problem 4

(a) $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$; then $A\mathbf{e}_1$, $A\mathbf{e}_2$, $A\mathbf{e}_3$ are its columns $(1,1)$, $(1,-1)$, $(3,0)$, as required.

(b) Kernel: $x + y + 3z = 0$, $x - y = 0 \Rightarrow x = y$, $z = -\frac{2}{3}x$: basis $\boxed{\{(3,3,-2)\}}$. Range: the columns $(1,1)$ and $(1,-1)$ are independent, so the range is all of $\mathbb{R}^2$, with basis $\boxed{\{(1,1),(1,-1)\}}$.

(c) The image is $\{x(1,1)+y(1,-1)+z(3,0) : 0 \le x, y, z \le 1\}$. The 8 corners map to $(0,0), (1,1), (1,-1), (3,0), (2,0), (4,1)$, the image is the filled hexagon with vertices $(0,0), (1,-1), (4,-1), (5,0), (4,1), (1,1)$ (the corners $(3,0)$ and $(2,0)$ land inside).

Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Midterm 2, Spring 2025 (Sara Billey) — Problem 5

With $O = \begin{bmatrix} -0.2 & 0.1 \\ 0.1 & 0.1 \end{bmatrix}$ (old grid vectors as columns) and $W = \begin{bmatrix} -0.5 & 0.3 \\ 0.7 & 0.4 \end{bmatrix}$ (new grid), a point is $O\mathbf{c}_{\text{old}} = W\mathbf{c}_{\text{new}}$.

(a)

$$\mathbf{c}_{\text{new}} = W^{-1}O\,\mathbf{c}_{\text{old}} = \boxed{\frac{1}{41}\begin{bmatrix} 11 & -1 \\ -9 & 12 \end{bmatrix}\mathbf{c}_{\text{old}}}.$$

(b) $\frac{1}{41}\begin{bmatrix} 11(20) - 40 \\ -9(20) + 12(40) \end{bmatrix} = \boxed{\left(\frac{180}{41},\ \frac{300}{41}\right) \approx (4.39,\ 7.32)}$.

Bonus. (a) *Always*: $\mathbf{0}$ is in both, and if $\mathbf{u}, \mathbf{w}$ are in both then so are $\mathbf{u}+\mathbf{w}$ and $c\mathbf{u}$. (b) Only when one contains the other ($U \subseteq V$ or $V \subseteq U$): otherwise pick $\mathbf{u} \in U \setminus V$, $\mathbf{v} \in V \setminus U$, and $\mathbf{u} + \mathbf{v}$ is in neither.